Evidence Routing and Supported Residuals for Cooperative LiDAR–Radar Perception

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Abstract

Cooperative perception depends on the quality of the local features that agents share. In adverse weather, missing LiDAR returns call for complementary evidence, whereas spurious returns call for suppression. We introduce DEER and SBR to separate these decisions from the support and amplitude of radar-based corrections. DEER learns independent missing and spurious scores from traceable point deletion and injection, then uses detached scores to route features while retaining each available modality. SBR selects local measured radar donors with a null option and bounds the resulting residual by observation support and current feature amplitude. Detection supervision learns the correction without a feature teacher. Seven configurations use a common 20-epoch, single-seed V2X-R protocol, with Snow held out from training and model selection. The joint model improves Snow Vehicle 3D average precision (AP3D) at intersection over union (IoU) 0.7 from 45.67 to 49.19 percent and the Fog/Snow mean by 1.67 percentage points, with Normal/Fog decreases of 0.26/0.19 points. A multiplicative type prior achieves the highest weather mean, 50.62 percent. The joint design is strongest at strict 3D overlap in held-out Snow, while type-dependent amplitude modulation improves average weather performance.

Keywords: cooperative perception, LiDAR–radar fusion, adverse weather, evidence routing, bounded residual